DATA 2x01 — Data Science, Big Data and Data Variety

Assignment number: 1 **Due Date:** May 18 2026, 11:59 PM
Weight: 20% **Assignment Type:** Group

Statistical Areas Level 2 (SA2s) [1] are medium-sized general purpose areas built up from whole Statistical Areas Level 1 (SA1s). Their purpose is to represent a community that interacts together socially and economically.

There are 2,473 SA2s covering the whole of Australia without gaps or overlaps. SA2s are generally the smallest areas used for the release of Australian Bureau of Statistics (ABS) non-Census of Population and Housing statistics, including Estimated Resident Population and Health and Vitals data.

SA2s generally have a **population** between 3000 and 25000 with an average of about 10000 people. SA2s in remote and regional areas generally have smaller populations than those in urban areas.

A **functional area** is the area from which people come to access services at a centre. This centre may be a rural town, a regional city, a commercial and transport hub within a major city, or the major city itself.

A centre and its functional area are represented by one or more SA2s. A rural town and its functional area may be combined into a single SA2. A larger town may be identified by a single SA2 and its functional area around the town by a second SA2. Larger towns and regional cities may be represented by several SA2s. Within cities, SA2s represent gazetted suburbs rather than functional areas.

SA2 coding structure An SA2 is identifiable by a 9-digit fully hierarchical code comprising the 1-digit State or Territory identifier, and Statistical Areas Level 2-4 identifiers. Then the SA2 identifier which is a 4-digit code, assigned in alphabetical order within an SA3. An SA2 code is only unique within a State or Territory if it is preceded by the State or Territory identifier. For example, the SA2 503021295 East Perth encodes the following information.

S/T	SA4	SA3	SA2	SA2 Name
5	03	02	1295	East Perth

The following figures capture the information discussed here [2].

[1] https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition jul2021-jun2026/main-structure-and-greater-capital-city-statistical-areas/

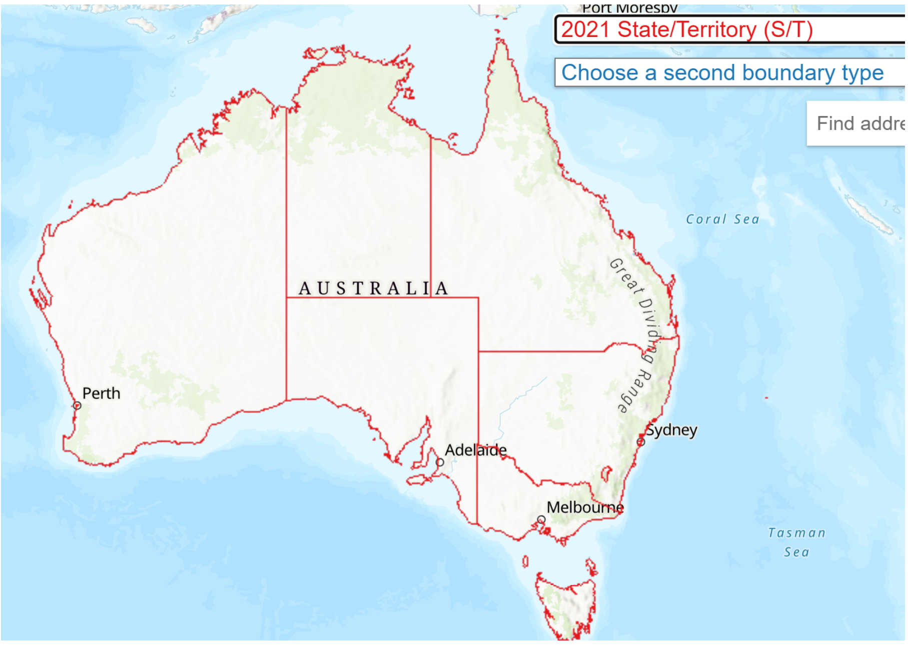

Figure 1: State/Territory

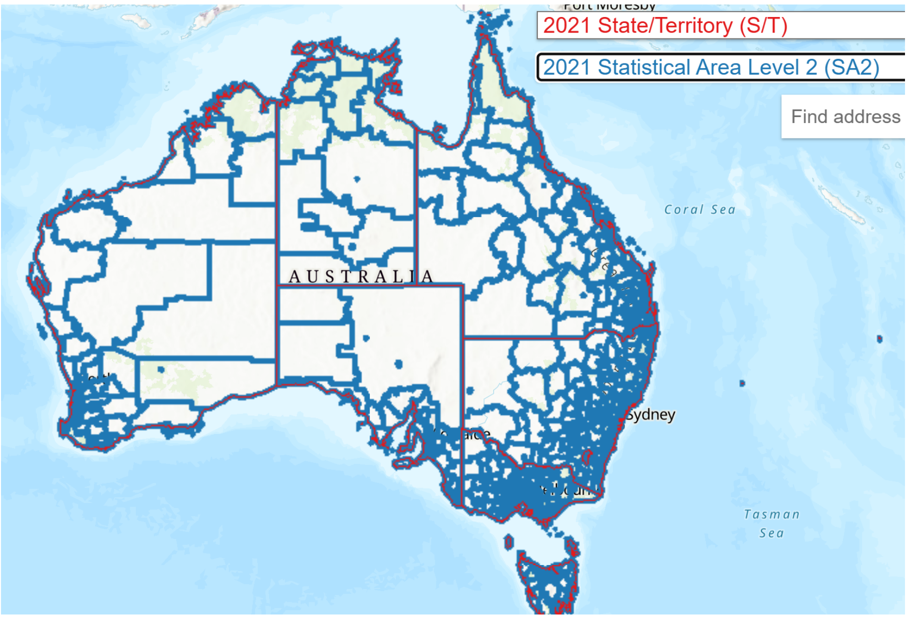

Figure 2: SA2s overlaid the States/Territories

statistical-area-level-2

[2]https://maps.abs.gov.au/

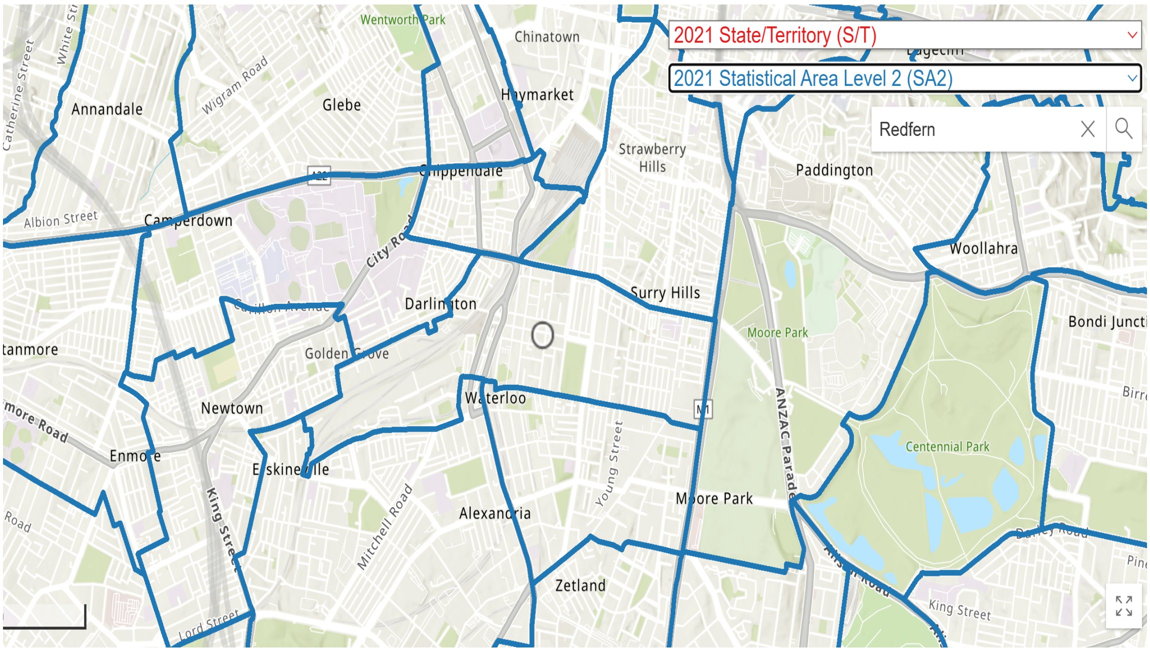

Figure 3: SA2 (zoomed in)

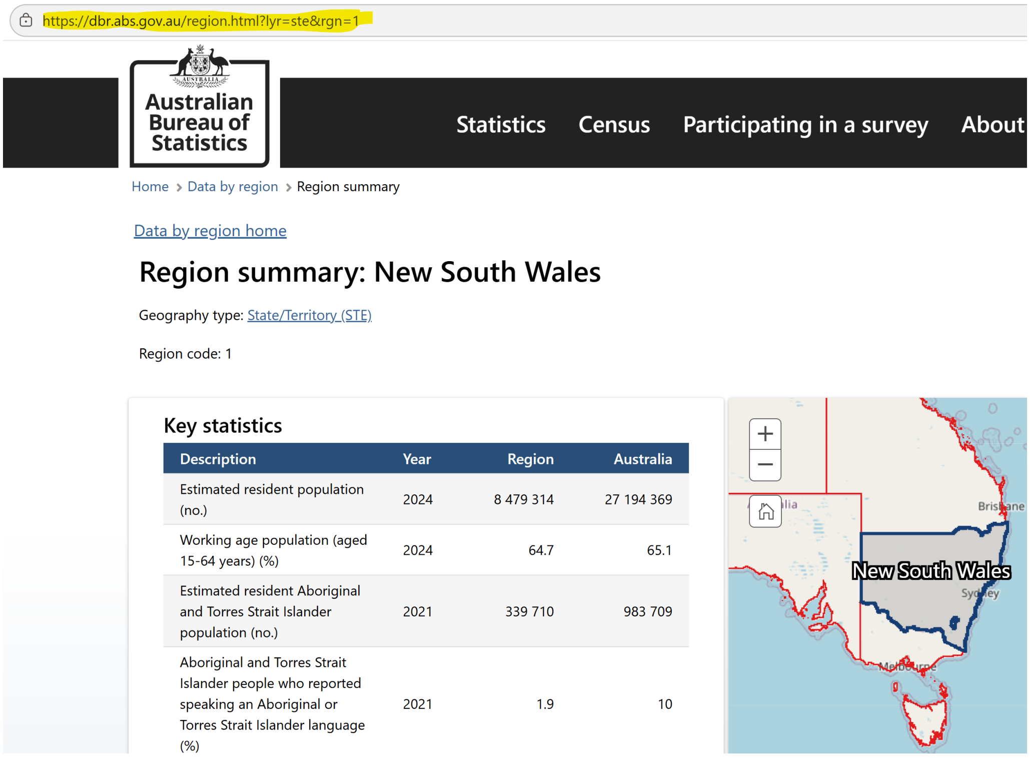

Figure 4: NSW statistics

Task-1 (30 points) The webpage `https://dbr.abs.gov.au/region.html?lyr=ste&rgn=1` (refer to Fig. 4) contain statistical information about New South wales.

- load the corresponding data (provided as the CSV file: Region_summary_NSW.csv) using Pandas. [5 points]
- perform data cleaning. [10 points]
- if your group has n members, then each member needs to separately come up with 5 (derived) statistics using Pandas. [15 points] (2 bonus points for coming up with interesting statistic(s)]

Task-2 (30 points) In Greater Sydney, there are 350+ SA2s, split between 15 SA4 zones (e.g. Northern Beaches, Parramatta). Each member in your group is to select one of these SA4 zones (i.e. a group of 4 people requires 4 distinct SA4 zones).

Utilise the **NSW Points of Interest API** to extract information relevant to each SA2 region and form a dataset:

- Develop a function that returns all points of interests from the API within a specified bounding box of coordinates (this can be done by adjusting the similar function in the Week 8 tutorial which finds all points of interest near a specified point). [5 points]

- Build a loop that cycles through each SA2 region within your selected SA4 region, runs the function for that region's bounding box, to find all points of interest within that SA2 region. [10 points]

- The cumulative results of this loop should then be ingested into your localhost database, as the final geographic dataset to be used in score calculations. Retain any columns of interest, use the **NSW Topographic Data Dictionary** as needed for data definitions, and ensure your table is well-defined accordingly. [15 points]

Task-3 (10 points) For every SA2 region in your selected zones, compute a score for how "well-resourced" they are according to the formula provided below, where S is the **sigmoid function**, z is the **normalised z-score**. Feel free to ignore any SA2 regions with a population below 100, if they exist amongst your chosen areas. You are welcome to extend the scoring function however you deem necessary, so long as rational explanation is provided (e.g., other mathematical standardisation techniques, mitigating the impact of outliers, etc).

$$\text{Score} = S\big(z_{\text{POI}}\big) \tag{1}$$

Task-4 (15 points) Prepare a report presenting thorough analysis of your results.

- Summarise key findings in Task 1; Interesting findings? (5 points)

- Visualise your scores in an engaging way (e.g. How were your scores distributed? What do they look like in a map-overlay visual?). (5 points)

- Scrutinise the results (e.g. What are the limitations of your scoring?). (5 points)

Task-5 (15 points) A brief conversation with your tutor (not a formal presentation) in the Week 12 tutorials (or Week 13, if necessary). This allows time to discuss the decisions behind your work, and **is a marked component, and is mandatory for any marks to be received.**

Additional Task (for DATA 2901 only)

- Using Twitter API, find out 100 recent tweets about Australia.

- Find the tweet with most likes.

Deliverables All deliverables are due (no later than) 11:59pm on Monday the 18th of May.

1. PDF Report

2. Jupyter Notebook describing your entire workflow

3. A brief conversation with your tutor

The **marking rubric** will be available on Canvas. **Late submission penalty**:−5% of the available marks per day late; minimum 0% after 5 days. Please submit **a single zip file** containing all deliverables electronically in Canvas, one for each group.

Participation As a group assignment, the mark awarded for your assignment is conditional on contribution to the group, and a baseline ability to explain the contents of your submission to your teaching team if asked. If members of your group do not contribute sufficiently, please alert your tutor as soon as possible. The tutor will have the discretion to scale the the mark received by an individual as below, based on the outcome of the group's demo.

Level of Contribution (in submitted work)	Proportion of Final Grade Received
No participation or no demo	0%
Passive member, but full understanding of the work	50%
Average contribution to the group's submission	75%
Major contribution to the group's submission	100%